

Lab Report 08

Lab Title: Implementation of Multi-Routing Protocol Network with DHCP

Objective

The objective of this lab is to design and implement a multi-network environment where:

- Three labs (Alpha, Beta, Charlie) are interconnected with a **core router**.
- Each lab uses a **different routing strategy** (Static Routing, RIP, OSPF).
- IP addresses are allocated using **DHCP** automatically.
- Finally, all networks are interconnected through the **core router**, which communicates routing information with the **ICT department**.

Network Topology

Lab Allocation: ID-56

- **Lab Alpha:** 192.168.56.0/24
- **Lab Beta:** 192.168.57.0/24
- **Lab Charlie:** 192.168.58.0/24

Configuration Rules per Lab

1. **Lab Alpha (Static Routing)**
 - 2 PCs connected.
 - PCs get IPs automatically from DHCP.
 - Router connected to core router using static routes.

2. Lab Beta (RIP Routing)

- 16 PCs connected via switches.
- Router runs **RIP v2** and shares routes with the core router.
- DHCP server assigns addresses from 192.168.57.0/24.

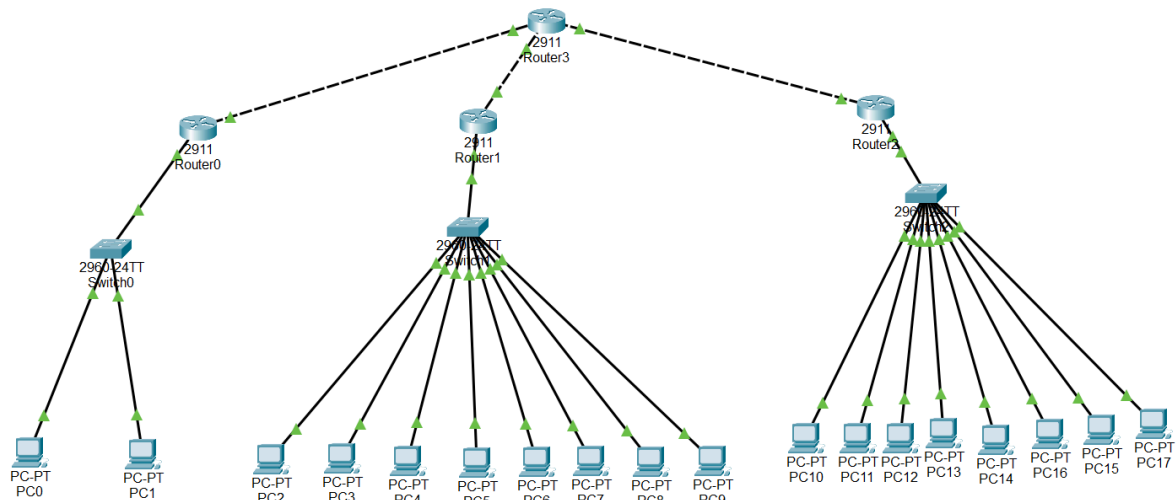
3. Lab Charlie (OSPF Routing)

- 16 PCs connected via switches.
- Router runs **OSPF** (single area, Area 0).
- DHCP server assigns addresses from 192.168.58.0/24.

4. Core Router

- Acts as the central hub.
- Connected to each lab router (Alpha, Beta, Charlie).
- Redistributes routes across static, RIP, and OSPF.

Network Topology



Routers:

- Alpha Router G0/0 → 192.168.56.1
- Beta Router G0/0 → 192.168.57.1
- Charlie Router G0/0 → 192.168.58.1
- Core Router G0/0 → 192.168.56.2
- Core Router G0/1 → 192.168.57.2
- Core Router G0/2 → 192.168.58.2

Lab Alpha (Static Routing)

1. Connect 2 PCs → Switch → Router (Alpha).
2. Configure DHCP pool on Alpha Router:
 - Network: 192.168.56.0/24
 - Default Gateway: 192.168.56.1
3. Configure PCs to obtain IP automatically.
4. Configure static routes on Alpha Router:
 - Route to 192.168.57.0 via Core Router.
 - Route to 192.168.58.0 via Core Router.

Lab Beta (RIP)

1. Connect 16 PCs → Switch → Router (Beta).
2. Configure DHCP pool on Beta Router:
 - Network: 192.168.57.0/24
 - Default Gateway: 192.168.57.1
3. Configure RIP v2 on Beta Router

4. Enable automatic route exchange with Core Router.

Lab Charlie (OSPF)

1. Connect 16 PCs → Switch → Router (Charlie).
2. Configure DHCP pool on Charlie Router:
 - Network: 192.168.58.0/24
 - Default Gateway: 192.168.58.1
3. Configure OSPF (Area 0)

Core Router Configuration

Redistributes routes across static, RIP, and OSPF:

```
router rip
version 2
network 192.168.57.0
redistribute ospf 1
redistribute static
!
router ospf 1
network 192.168.58.0 area 0
redistribute rip
redistribute static
```

- Configures static routes to Alpha Network.
- Sets default route to ICT Department Router.

DHCP Configuration Example (on Lab Alpha Router)

```
ip dhcp pool ALPHA_POOL
network 192.168.56.0 255.255.255.0
default-router 192.168.56.1
```

DHCP would be set similarly for the other labs as well.

Pings

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC10	PC9	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC9	PC10	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	3	(edit)	(delete)

Challenges

- Route redistribution was a little bit challenging.